Equality in Tuza's Inequality for 5-Degenerate Graphs

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Abstract

We characterize equality in Tuza's triangle packing–covering inequality for finite simple 5-degenerate graphs. Equality holds exactly when every nonempty triangle component, connected through shared graph edges, has edge support K_4 or K_5. We prove the combined statement, including $\tau(G) \le 2\nu(G)$, where $\nu(G)$ is the maximum number of edge-disjoint triangles and $\tau(G)$ is the minimum number of graph edges meeting every triangle. Triangle-free background edges and clique blocks sharing vertices are allowed; graph order and maximum degree are unrestricted. The inequality has a short deduction from published local reductions. The equality classification requires additional residual-packing compatibility and a strict-cover saving for arbitrary outside attachments. The exact combined theorem has been checked in Lean with pinned dependencies. Historical priority and external specialist peer review remain unestablished.

Introduction and precise contribution

The equality classification is the principal contribution proposed in this report. Tuza's conjecture [1] asks whether $\tau(G) \le 2\nu(G)$ holds for every finite simple graph. We establish the inequality together with its exact equality characterization in the subgraph-closed class of 5-degenerate graphs. The inequality admits a short deduction using published local reductions, given in Appendix A; a short proof does not by itself establish that the resulting inequality was previously known. Historical priority remains unestablished.

Theorem 1 (Main theorem). Every finite simple 5-degenerate graph G satisfies

$$\tau(G) \le 2\nu(G).$$

Equality holds if and only if every nonempty shared-edge triangle component has edge support K_4 or K_5.

A graph is **5-degenerate** if every nonempty vertex-induced subgraph has a vertex of degree at most five in that subgraph. Equivalently, every nonempty subgraph has such a vertex: deleting edges cannot increase degrees. This is a degeneracy condition, with no bound on original maximum degree. It does not imply maximum average degree less than seven. The theorem includes the empty graph and every triangle-free graph: both parameters vanish and the component condition is vacuous.

Two triangles are adjacent if they share an **edge**. Their connected components under chains of this adjacency are the *triangle components*. The edge support of a component is the union of its triangle edges. This differs from ordinary graph connectedness: distinct triangle components may share vertices, and edges lying in no triangle affect neither parameter. All triangle families used below are the full families of their graphs unless a subfamily is expressly named. A string abc denotes the triangle on distinct vertices a, b, c, and ab its graph edge. We write $N(v)$ and $N[v]$ for open and closed neighborhoods, $d(v)$ for degree, and $G[N(v)]$ for a vertex link.

Equivalent block formulation. The component condition is equivalent to the existence of a finite family of pairwise edge-disjoint actual K_4 and K_5 subgraphs containing every graph triangle. To prove this, component supports give such a family directly. Conversely, assign each triangle to a containing block. Triangles in different blocks cannot share an edge, by edge disjointness. Within K_4 or K_5 its triangles are connected by shared edges: two triangles sharing only a vertex are joined by a triangle using that vertex and one other vertex from each. Every block edge belongs to a triangle. Thus a triangle component consists exactly of one block's full triangle family, and its literal edge support is that clique. This argument checks the literal component condition rather than assuming it from a supplied decomposition. Since packing and covering numbers add over edge-disjoint triangle supports, the elementary parameters in Lemma 2 prove the equality converse for arbitrary graphs.

The proof's starting framework is due to Puleo [2, Definition 2.1 and Lemma 2.2]. His robustness and local low-degree results [2, Lemmas 2.6–2.7 and Propositions 5.1–5.2] are predecessors of Sections 3–6; the four/eight reducing pair in Section 6.2 is the witness in his Proposition 5.2, Case 1. Botler, Fernandes and Gutiérrez [3, Lemma 3.3] provide the direct predecessor of Proposition 8 for irreducible robust graphs. The extra requirement here is that balanced reductions need not contradict a minimal failure of the **combined** statement. We therefore derive exact residual equality, classify its clique supports, and prove compatibility and strict improvements at each balanced case. These extensions, especially arbitrary outside-page compatibility and the thirteen-versus-fourteen cover saving, remain explicit below. Appendix A gives the shorter credited inequality deduction, for which balanced reductions suffice.

Cui, Haxell and Ma [4, Theorem 1.2 and Section 2] characterize planar equality using edge-disjoint K_4 copies containing every triangle, with triangle-free background edges allowed. Theorem 1 recovers their known case because planar graphs are 5-degenerate and contain no K_5. Their theorem is not claimed as a new consequence in historical terms. Puleo's main sparsity theorem assumes maximum average degree less than seven. This class and the class of all five-degenerate graphs are incomparable. The graph K_7 has maximum average degree six but is not five-degenerate. In the other direction, start with K_5 and add five pairwise nonadjacent vertices, each joined to every original clique vertex. There are ten vertices and $10 + 25 = 35$ edges, so the average degree is seven and the maximum average degree is at least seven. Removing the five new degree-five vertices first and then the clique gives a five-degeneracy ordering. Thus this graph is outside Puleo's hypothesis. We use his local framework rather than a containment between these classes. Botler–Fernandes–Gutiérrez prove a treewidth theorem; their local lemma is the antecedent used here, not an invented degeneracy version of their main result.

Recent context. Gupta [5, Theorem 1.3], arXiv:2608.06538v1 (6 August 2026), proves Tuza's inequality for graphs of maximum degree at most seven. His Question 12.3 asks for the equality cases in that class. Theorem 1 supplies the equality classification on its intersection with the five-degenerate class. It does not answer the whole question: in particular, a nonempty seven-regular graph is not five-degenerate. Gupta's maximum-degree theorem is not used to prove Theorem 1.

Liu and Zeng [6, Theorem 1.3], arXiv:2610.15539v1 (11 October 2026), establish the inequality for every fixed $c < 2$ and sufficiently large n under $e(G) \geq n^2/4 - cn$. This is an edge-count condition, distinct from the degeneracy assumption here; no density-class conclusion is inferred from Theorem 1. The separate triangle-restoration report [8] retains its own statements and verification boundary.

Reading the proof

Sections 1–5 establish clique compatibility, tight criticality, robustness, and the two possible degree-five links. Sections 6–7 prove all degree-five adjacency, neighbor and overlap restrictions for a minimal combined obstruction. Section 8 applies degeneracy a second time, reconstructs the final core, and excludes arbitrary outside attachments. Section 9 closes the induction. Section 10 gives a concise verification account. Appendix A contains the short inequality proof, Appendix B the small-link proof organization used in Lean, and Appendix C the detailed declaration coverage and verification provenance. Every ordinary proof needed for Theorem 1 is contained here.

For a local vertex set W, an **external page** is a triangle with base edge in W and apex outside the specified core; its apex-to-base edges are its spokes. Different pages may share an apex. A *surviving footprint* means only edges actually remaining in the explicitly named residual graph. No bound is imposed on outside vertices or edges. Partial spoke deletion at a closed link component removes just its spokes, retaining all other neighbors; closure ensures that no unlisted hub triangle crosses to another link component.

1. Elementary block facts

Lemma 2 (Clique parameters and compatibility). The graphs K_4 and K_5 have packing/covering numbers (1,2) and (2,4). A maximum K_4 packing avoids a star; a maximum K_5 packing avoids any two prescribed edges. Every edge belongs to some minimum cover.

Proof.

K_4 has $(\nu, \tau) = (1, 2)$. K_5 has $(\nu, \tau) = (2, 4)$: two edge-disjoint triangles exist; three edge-disjoint triangles would use nine edges and leave one edge, impossible because every degree in a union of edge-disjoint triangles is even whereas removing one edge from K_5 creates two odd degrees. Each edge of K_5 meets three of its ten triangles, so three cover edges cannot cover them all; four edges inside a vertex bipartition of sizes two and three do cover all triangles.

A K_4 maximum packing can avoid any star of forbidden edges by choosing the face opposite the star center. A K_5 maximum packing can avoid any two edges. Up to relabeling, for forbidden adjacent edges 12,13 choose 145,234; for forbidden disjoint edges 12,34 choose 135,245. Each displayed pair is edge-disjoint. Zero or one forbidden edge follows by enlarging the forbidden set.

In either block there is a minimum cover containing any specified edge. In K_4 take it and its opposite edge. In K_5 take the four edges inside a $2 + 3$ vertex partition whose two-vertex class is that edge's endpoints. Independent choices combine across triangle components, whose edge supports are disjoint.

We also use the elementary intersecting-triangle lemma: a pairwise edge-intersecting triangle family has a two-edge cover; if it has no common edge, all its triangles are faces of a single K_4. Indeed, two distinct triangles can be written *abc*,*abd*. A triangle avoiding *ab* must share another edge with each, hence is *acd* or *bcd*. Once such a triangle is present, every further triangle lies among those four faces. The pair *ab*,*cd* covers the family. Families with a common edge, or a single

triangle, are immediate.

Lemma 2A (Block-packing compatibility). Let K be a subgraph of G whose full triangle components have K_4 or K_5 supports, and let P be a family of k pairwise edge-disjoint G-triangles. For each component block B, let $D_B = E(B) \cap \bigcup_{T \in P} E(T)$. Suppose D_B is contained in a star when B is K_4, and $|D_B| \leq 2$ when B is K_5. Then $\nu(G) \geq \nu(K) + k$.

Proof. By Lemma 2 choose a maximum packing in each block avoiding D_B. The blocks have disjoint edge supports, so these packings combine into a packing of size $\nu(K)$. It is edge-disjoint from every member of P by the definition of D_B. Appending P proves the bound.

Two-page use and its bound. If a terminal set S is independent in K, each clique block meets S in at most one vertex. For two pages with bases in S, each page contributes at most one surviving edge to a block: a block containing both spokes would contain both base endpoints. The union of the surviving spokes is therefore a star with at most two edges, satisfying both cases of Lemma 2A. A star alone is not sufficient for a K_5 block. In every application below we state the bound on the entire surviving footprint, including any contribution from local packed triangles.

2. Minimal-obstruction framework in a hereditary class

Lemma 3 (Tight criticality). An edge-minimal failure of the combined statement in a subgraph-closed class, after removing isolated vertices, has one nonempty triangle component, no triangle-free edges, and $\tau(G) = 2\nu(G)$. Every edge deletion preserves ν and decreases τ by one.

Proof.

Call S the assertion that Tuza's inequality holds and equality implies the K_4/K_5 component condition. The converse has already been proved in the introduction. A failure therefore means either $\tau(G) > 2\nu(G)$, or $\tau(G) = 2\nu(G)$ with an unclassified component. Let C be any class closed under taking subgraphs, and suppose G is a failure of S in C with as few edges as possible. Discard isolated vertices. Then every proper edge-subgraph satisfies S.

There are no triangle-free edges: deleting one preserves the triangle family and the failure. There is only one nonempty triangle component. Indeed, the support subgraph of a triangle component has exactly that component as its full triangle family: any triangle using its support edges shares an edge with a component member, so belongs to the same shared-edge component. If there were several nonempty components, each support subgraph would have fewer edges and minimality would apply S to it. Summing their inequalities, equality of the sum forces equality and the block classification in every component. Thus G has connected nonempty triangle support.

Write $h = \nu(G)$. A failure of S has $\tau(G) \geq 2h$. If $\nu(G - f) = h - 1$ for an edge f, induction gives $\tau(G) \leq \tau(G - f) + 1 \leq 2h - 1$, impossible. Hence every edge deletion preserves ν.

If $\tau(G - f) = 2h$ for $f = xy$, then $G - f$ has only K_4/K_5 triangle components. Choose a triangle xyw of G. The wedge edges xw,yw cannot lie in the same residual clique block, since xy is absent. Each residual block contains at most one of xw, yw because xy is absent. Thus its forbidden footprint is a star of size at most one. Lemma 2A, with $P = \{xyw\}$, gives $h+1$ triangles, contradiction. Therefore $\tau(G - f) \leq 2h - 1$ for every edge f. Cover deletion changes τ by at most one, so

$$\tau(G) = 2h; \nu(G - f) = h; \tau(G - f) = 2h - 1 \quad \text{for every edge } f$$

Thus every minimal S-obstruction in a hereditary class is tight-critical. This argument proves

the reduction; it does not presume global Tuza.

Lemma 3A (Balanced-reduction accounting). Let G be the edge-minimal combined obstruction of Lemma 3, write $h = \nu(G)$, and let K be a proper subgraph in the same subgraph-closed class. Suppose that P consists of r pairwise edge-disjoint G-triangles, no edge used by P lies in the triangle support of K, and $X \subseteq E(G)$ meets every G-triangle not belonging to the full triangle family of K. If $|X| \leq 2r$, then

$$\nu(K) = h - r, \qquad \tau(K) = 2\nu(K), \qquad |X| = 2r,$$

and K has the component-block classification. In particular, $|X| < 2r$ is impossible.

Proof. The support condition lets every K-packing append P, so $\nu(K) + r \leq h$. A minimum K-cover together with X covers every G-triangle by the stated lost-triangle condition. Minimality supplies both the inequality and the equality conclusion for K. Consequently

$$2h = \tau(G) \leq \tau(K) + |X| \leq 2\nu(K) + 2r \leq 2h.$$

Every inequality is an equality. This gives the displayed accounting and licenses the residual component classification. No residual equality is assumed before this chain is proved.

A useful instance is the reducible-set framework of Puleo [2, Definition 2.1 and Lemma 2.2]. For a nonempty vertex set U, a packing P of size r, and an edge set X hitting every triangle incident with U and containing every P-edge with neither end in U, put $K = (G - X) - U$. The packing and lost-triangle hypotheses of Lemma 3A then follow directly. The partial-spoke deletions below use the same hypotheses, checked on their displayed witnesses.

3. An edge cannot belong to just one triangle

Lemma 4 (Triangle codegree). Every edge of a minimal combined obstruction belongs to at least two triangles.

Proof.

Suppose uv belongs only to uvw. Delete uw,vw to get K. The edge uv is triangle-free in K, so every K-packing can be appended by uvw. A K-cover plus uw,vw covers G. Hence $\nu(K) = h - 1$, $\tau(K) = 2h - 2$ and K has the block classification. Each block meets $U_0 = \{u, v, w\}$ at most once: uw,vw are absent, and uv is triangle-free in K.

Let F be the full family of triangles lost in this deletion. They all contain w. If F is intersecting, the preceding lemma says that F has a common edge or is contained in K_4 faces. In the latter case, since uvw belongs to F and all F-triangles contain w, absence of a common edge forces wuv,wux,wvx for a fourth vertex x. Their edges force the triangle uvx, contradicting uniqueness of uvw. Thus F has a one-edge cover, giving $\tau(G) \leq \tau(K) + 1 < 2h$.

If F is not intersecting, take two disjoint members. Each block meets U_0 in at most one vertex. The two selected triangles therefore have a star footprint there, with each triangle contributing at most one surviving edge. Its size is at most two, including for K_5. Lemma 2A gives $h + 1$ triangles, contradiction. Therefore every G-edge has triangle codegree at least two. Equivalently every vertex in every link $G[N(v)]$ has degree at least two.

4. Every link component has at least five vertices

Lemma 5 (Robustness). Every connected component of every vertex link in a minimal combined obstruction has at least five vertices; in particular the obstruction G has minimum degree at least five.

Proof.

Let C_0 be a connected component of the link at v. In the following cases, deleting v means deleting just the spokes into C_0; other neighbors of v remain. There is no triangle through a deleted spoke with its third vertex outside C_0, because that would join link components. All listed covers therefore account for every lost triangle. External rim pages have apex outside $C_0 \cup \{v\}$.

4.1 Link component K_3

Write $C_0 = abc$. Delete its three v-spokes and ab. The triangle vab appends to any residual K-packing, and $\{ab, vc\}$ is a cover increment, so K has equality blocks and $\nu(K) = h - 1$.

An external abx combines with vbc. Their edges present in K are bc,ax,bx. Since ab is deleted, no block contains both a and b; the only possible two-edge overlap is bc,bx, a star at b. The whole footprint in a block is thus a star of size at most two; Lemma 2A yields $h + 1$ triangles, contradiction. By symmetry no rim edge of abc has an external triangle. Thus the K_4 on v,a,b,c is an isolated triangle component. Connectedness makes it the whole triangle support, which satisfies S, contradiction.

4.2 Link component K_4

Write $C_0 = abcd$. Delete its v-spokes and ab,cd. The packing vab,vcd appends, and $\{ab, cd, va, vb\}$ is a four-edge cover increment. Hence $\nu(K) = h - 2$ and K has equality blocks.

An external abx combines with vac,vbd. Its old forbidden edges are ac,bd,ax,bx. Because ab is absent, a block sees at most two of these edges, necessarily adjacent at a or b. The whole footprint is a star with at most two edges, so Lemma 2A gives $h + 1$ triangles. Varying the perfect matching ab,cd excludes external triangles on every rim edge. The K_5 on v,a,b,c,d is therefore an isolated triangle component, again impossible.

4.3 Diamond link component

Write $C_0 = K_4 - ab$ on a,b,c,d. Delete its v-spokes and ac,bd. Use packing vac,vbd and cover increment $\{ac, bd, vc, vd\}$; equality gives $\nu(K) = h - 2$ and blocks.

An external acx combines with vad,vbc. Its old forbidden edges are ad,bc,ax,cx. Since ac is absent in K, a block sees at most two of these, with the only possible pairs ad,ax or bc,cx. Each full block footprint is a star of size at most two; Lemma 2A contradicts maximality. By symmetry, including use of the other matching ad,bc, no external triangle uses ac,ad,bc,bd. Returning to K above, a K-cover plus $\{vc, vd, cd\}$ covers G: vc,vd cover all v-triangles, and the only other lost triangles are acd,bcd, hit by cd. This gives $2h - 1$ cover edges, contradiction.

4.4 Four-cycle link component

Let C_0 be the cycle *abcd*. Delete its v-spokes and *ab*,*cd*,*da*, retaining *bc*. The packing *vab*,*vcd* appends; all four rim edges are a cover increment. Thus $\nu(K) = h - 2$, $\tau(K) = 2h - 4$ and K has equality blocks.

By Section 3, *bc* has a triangle *bcx* other than *vbc*. It survives in K: x cannot be a or d since the diagonals *ac*,*bd* are absent. Hence *bc* lies in a residual block. Choose a minimum K-cover containing *bc*, and add only *ab*,*cd*,*da*. It covers G with $2h - 1$ edges, contradiction.

Since every link has minimum degree at least two, its connected components of order at most four are exactly K_3,C_4,diamond,K_4. All are excluded. Hence every link component has order at least five, and $\delta(G) \geq 5$.

5. Complete classification of five-vertex links

Lemma 6 (Degree-five links). The link of every degree-five vertex in a minimal combined obstruction is K_5 or K_5 minus one edge.

Proof.

Robustness forces the link of a degree-five vertex to be connected: its components each have at least five vertices, and its total order is five. We first exclude its 3-colorable possibilities, and then treat the non-3-colorable case.

Suppose a link component L has five vertices, minimum degree at least two, and is 3-colorable. Its complement has maximum degree at most two. A 3-coloring gives either a size-three color class (a triangle in the complement) or two size-two classes (a matching of size two). There are eleven unlabeled graphs on five vertices of maximum degree at most two: $5I$, $K_2 + 3I$, $2K_2 + I$, $P_3 + 2I$, $P_3 + K_2$, $P_4 + I$, P_5, $C_3 + 2I$, $C_3 + K_2$, $C_4 + I$, and C_5. Here $I = K_1$ and $+$ denotes disjoint union. The first, second, and fourth entries have non-3-colorable complements; the other eight are the possible complements here. Thus the possible complements are exactly

$$2K_2 + I, P_3 + K_2, P_4 + I, P_5, C_3 + K_2, C_3 + 2I, C_4 + I, C_5$$

For a matching M and vertex set Q with $L - Q$ a subgraph of M, delete the local v-spokes and M. The $|M|$ triangles at v on M append to every residual packing, while M plus the v-spokes to Q is a cover increment. If $|Q| < |M|$ this is strict; if equality holds, the residual has K_4/K_5 blocks and packing $h - |M|$.

5.1 Four dense types

In each dense case $|M| = |Q| = 2$. The listed three triangles are mutually edge-disjoint. Their edges present in the residual are listed as their surviving footprint. The missing and deleted edges restrict a block's intersection as shown, so every whole block footprint is a star with at most two edges. Lemma 2A appends all three triangles, contradiction.

Dense case 1. Let L be $K_{2,3}$ with sides $\{p, q\}$,$\{b, c, x\}$, plus *bc*. Take $M = \{bc, px\}$ and $Q = \{p, q\}$. The three triangles are *pbc*; *vqb*; *vpx*; their surviving footprint is $\{pb, pc, qb\}$. Possible two-edge overlap in a block: none: *bc* deleted and *pq* absent.

Dense case 2. Let L be K_5 minus *ab*,*bc*,*cd*; fifth vertex e. Take $M = \{bd, ce\}$ and $Q = \{a, e\}$. The three triangles are *ace*; *vad*; *vbe*; their surviving footprint is $\{ac, ae, ad, be\}$. Possible two-edge

overlap in a block: only ae,ad.

Dense case 3. Let L be K_5 minus ab,bc,cd,de. Take $M = \{ac, bd\}$ and $Q = \{a, e\}$. The three triangles are ace; vad; vbe; their surviving footprint is $\{ae, ce, ad, be\}$. Possible two-edge overlap in a block: none.

Dense case 4. Let L be $K_{2,3}$ with sides $\{p, q\}$,$\{a, b, c\}$, plus pq. Take $M = \{ap, bq\}$ and $Q = \{p, q\}$. The three triangles are apq; vbp; vcq; their surviving footprint is $\{aq, pq, bp, cq\}$. Possible two-edge overlap in a block: only pq,cq.

Both allowed pairs are adjacent. For completeness, in dense case 2 ac with ae requires deleted ce; ac with ad requires absent cd; ac with be requires absent ab; ae with be requires absent ab; ad with be requires deleted bd. Only ae,ad remains, precluding a triple. In dense case 3, the pairs respectively require deleted ac, absent de, absent ab, deleted ac, absent bc, or deleted bd. In dense case 4, every pair except pq,cq requires deleted ap or bq, or an absent pair among a,b,c. Dense case 1's three pairs require bc or pq. These are all pair tests.

5.2 Three sparse types

For a bowtie $L = abc \cup ade$, take $M = \{bc, de\}$, $Q = \{a\}$. The cover increment has size three for a compatible two-packing, immediately strict.

For $L = K_{2,3}$ with sides $\{a, b\}$,$\{c, d, e\}$, take $M = \{ac, bd\}$, $Q = \{a, b\}$. An external acx combines with vad,vbe. The old forbidden edges are ad,be,ax,cx. Missing/deleted ac,ab,cd,ce exclude every pair except ad,ax, adjacent at a. The footprint has at most two edges in each block and is a star, so Lemma 2A excludes an external ac-page; symmetry handles every rim edge. There are no triangles within L, so each rim edge belongs only to its v-triangle, contradicting Section 3.

For $L = C_5$ with rim $abcde$, take $M = \{ab, cd\}$, $Q = \{c, e\}$. An external abx combines with vbc,vde. Old forbidden edges are bc,de,ax,bx. Deleted ab,cd and absent ad,bd leave only the possible pair bc,bx. The footprint in each block is a star of size at most two. Lemma 2A excludes such an external page, and symmetry excludes all rim pages. Again each rim edge belongs to just its v-triangle, contradiction.

5.3 The remaining type $K_{2,2,1}$

The local four-triangle/eight-edge witness is Puleo's [2, Proposition 5.1, proof Case 1]; the residual-equality and external-page compatibility extension is proved here.

Write the missing link edges ab,cd and the fifth link vertex e. The local graph B on $W = \{v, a, b, c, d, e\}$ is K_6 minus ab,cd. Delete all B-edges to K. The four edge-disjoint triangles vac,vbd,ade,bce append to every K-packing. All eight rim edges cover every lost triangle through v and every other lost triangle. Hence equality forces $\nu(K) = h - 4$ and $\tau(K) = 2h - 8$, and K has equality blocks. Each block meets W at most once, since W is independent in K.

Let F be the external rim pages, triangles with a rim edge and apex outside W. The lost triangles are exactly the B-triangles and F. B has the four-edge cover $\{va, vb, ec, ed\}$; therefore equality forces $\tau(F) \geq 4$.

Call the following four pairs of rim edges bad:

$$\{ac, bd\}, \{ad, bc\}, \{ae, be\}, \{ce, de\}$$

These four bad pairs form a perfect matching on the eight rim edges: every rim edge has exactly

one bad partner. A third, distinct base is therefore good with both members of a bad pair.

Every other pair f,g is good and $B - \{f, g\}$ has three edge-disjoint triangles. The symmetries swapping a,b, swapping c,d and exchanging the two two-vertex parts reduce all good pairs to four cases:

f,g	Three triangles avoiding them
ac,ad	vae; bce; vbd
ae,ce	bde; vad; vbc
ac,ae	bde; vad; vbc
ac,be	ade; vbd; vce

The four cases are adjacent pairs of cycle edges, spokes at e to different parts, an incident mixed pair, and a nonincident mixed pair. The remaining cycle pairs and spoke pairs are precisely those listed as bad. Thus the list covers every good pair without an appeal to enumeration.

If F has no edge-disjoint pair on good bases, then $\tau(F) \leq 2$. To prove this, an intersecting F already has a two-edge cover. Otherwise choose disjoint pages on bases f,g, which must be bad partners. Every page on a third base must intersect both, since a third base is good with each bad partner. Such intersections require a common external apex and bases meeting both f and g.

For adjacent bad partners ae,be (or ce,de), disjointness requires distinct apices, so no third base occurs; f,g cover F. The same holds for disjoint partners with distinct apices. For disjoint partners ac,bd with common apex x, a third base meeting both is only ad or bc, and its page has apex x. Its presence forces every page on ac or bd to have apex x, or else it would form a good disjoint pair with that third page. The other possible third base likewise has apex x. Thus F lies in the four pages at x over the cycle ac,cb,bd,da, covered by xa,xb. If there is no third base, ac,bd cover F. This proves the claim; symmetry handles ad,bc.

Since $\tau(F) \geq 4$, F has a disjoint pair on good bases f,g. Combine these two pages with the displayed local three-packing avoiding f,g. Each residual block meets W at most once, and hence meets each external page in at most one old spoke. If it meets both, the forbidden spokes share its unique W vertex. The local three-packing has no surviving edges because all B-edges were deleted. Thus the entire footprint is the indicated star of at most two spokes, including for K_5. Lemma 2A appends all five selected triangles, yielding $h + 1$, contradiction.

This completes exclusion of every 3-colorable five-vertex link component.

5.4 One additional non-3-colorable link

The link consisting of K_4 on a,b,c,d and a fifth vertex x adjacent exactly to a,b is also excluded. Take $M = \{cd, ax\}$ and $Q = \{a, b\}$; $L - Q$ is the single edge cd. Deleting the local v-spokes and M forces equality blocks with packing $h - 2$. The triangles acd,vax,vbd are edge-disjoint and have old forbidden edges ac,ad,bd. A block cannot contain both ac,ad or both ac,bd because cd is deleted; the only possible pair is ad,bd, adjacent at d. The entire footprint is consequently a star with at most two edges. Lemma 2A appends the three triangles, giving $h + 1$, contradiction.

5.5 The two remaining degree-five links

Every K_4-free graph on at most five vertices is 3-colorable. Indeed, a vertex-minimal graph that is not 3-colorable has minimum degree at least 3. On four vertices it is K_4; on five its complement

is a matching. K_5 minus zero or one edge contains K_4, while K_5 minus two independent edges is $K_{2,2,1}$ and is 3-colorable.

Thus a degree-five link not already excluded contains K_4. Its fifth vertex has at least two neighbors there. Exactly two is the configuration excluded in Section 5.4; three gives K_5 minus one edge, and four gives K_5. Hence every degree-five vertex of a minimal S obstruction has one of these two links.

6. Degree-five vertices are independent and have no degree-six neighbor

Lemma 7 (Low-neighbor restriction). Degree-five vertices form an independent set in a minimal combined obstruction, and every neighbor of such a vertex has degree at least seven.

Proof.

Attribution and extension. The following packing/cover accounting is Puleo's reducible-set framework (Definition 2.1, Lemma 2.2). His Proposition 5.2 is the antecedent for low-degree adjacency, including the precise balanced witnesses in Section 6.2. For the combined obstruction, only a strict reduction is an immediate contradiction. Every balanced case below explicitly supplies the additional equality argument.

We use the reducible-set instance of Lemma 3A with the actual packing and cover witnesses displayed in each case. A strict cover bound contradicts that lemma; a balanced bound yields its exact residual equality and classification.

The singleton matching version takes a matching M in a link and a vertex set Q such that the link minus Q is a subgraph of M. Delete the central vertex and M. The hub triangles on M give $|M|$ packed triangles, and M plus the spokes to Q give a cover increment $|M| + |Q|$. The set Q may contain endpoints of edges in M.

Residual-support observation. In each application, the residual graph K has a five-vertex set R such that $K[R]$ is K_5 with one edge or two independent edges deleted. These deleted pairs are nonedges of K. In each such induced graph all remaining edges lie in one triangle component with five vertices. It cannot lie in a K_4 block; it cannot lie in a K_5 block because the specified missing edge was deleted from the whole residual. This can also be seen by listing their triangles or starting with two overlapping K_4 copies in the one-edge case.

6.1 Dominating neighbors of degree five or six

Let v have degree 5 and let u be a neighbor with $N[v] \subseteq N[u]$. Write $W = N(v) - \{u\}$, $B = G[\{u, v\} \cup W]$, and $Z = N(u) - N[v]$. Then B is K_6 or K_6 minus one rim edge, and $|Z| = d(u) - 5$.

Take $X = E(G[W]) \cup \{uv\} \cup \{uz : z \in Z\}$. It meets every triangle at u or v. If B is $K_6 - ab$ and $|Z| \leq 1$, X has at most 7 edges and B has the four-packing auv,acd,bud,bcv ($W = \{a, b, c, d\}$); this is strict. If $B = K_6$ and Z is empty, $|X| = 7$ and again a four-packing is strict.

If $B = K_6$ and $Z = \{p\}$, the edge up belongs to a triangle, so p has a neighbor w in W (v has no neighbor p). Choose a four-packing of K_6 avoiding uw and add upw. Add pw to X. Its size is $9 < 10$, strict. The required four-packing exists because K_6 minus a perfect matching decomposes into four triangles: for nonedge pairs ab,cd,st, use acs,adt,bct,bds; extend uw to such a matching.

Thus no dominating neighbor of a degree-five vertex has degree 5 or 6.

6.2 Adjacent degree-five vertices

If two degree-five vertices u,v are adjacent and have four common neighbors, they dominate each other and Section 6.1 applies. Otherwise, their precise links force three common neighbors $W = \{w_1, w_2, w_3\}$. Both links are K_5 minus one edge. Let p be the unique neighbor of u outside $N[v]$ and q the unique neighbor of v outside $N[u]$. The vertices p,q are distinct, each joins all of W, vp and uq are absent, and W is a triangle. Whether pq is present is irrelevant.

Use

$$P = \{uw_1w_2, vw_1w_3, upw_3, vqw_2\},$$

$$X = E(W) \cup \{uv, up, vq, pw_3, qw_2\}$$

This is a balanced four/eight reduction deleting u,v. Its residual K has equality blocks and W is independent there. The five triangles

$$uvw_1, upw_2, vqw_3, pw_1w_3, qw_1w_2$$

are edge-disjoint. Their surviving edges in K are exactly pw_1,pw_2,qw_1,qw_3. A residual block meets W at most once, so its whole surviving footprint has at most two edges sharing that vertex. Lemma 2A appends all five triangles, contradiction. Hence the degree-five vertices form an independent set.

6.3 A nondominating degree-six neighbor

Suppose v has degree 5 with link $K_5 - ut$, and u is one of the missing-edge endpoints. Thus $N(v) = \{u, t, a, b, c\}$, ut is absent, and $W = \{a, b, c\}$ is a triangle. If $d(u) = 6$, write its two remaining neighbors p,q outside $N[v]$. In $G - u$, the vertices $\{v, t, a, b, c\}$ induce K_5.

If pq is present, in the link at u take $M = \{pq, va, bc\}$ and $Q = W$. Outside Q the only link edge is pq. This balanced three/six singleton reduction leaves that K_5 minus the two independent edges va,bc, impossible for residual equality blocks.

If pq is absent, triangle codegree at least 2 for up and uq implies that p,q each have at least two neighbors in W. Choose distinct $a \in N(p) \cap W$ and $b \in N(q) \cap W$, with c the third vertex. Take $M = \{pa, qb, vc\}$ and $Q = W$. The link outside Q is independent. This balanced three/six reduction leaves the K_5 with only vc deleted, again impossible.

Together with the minimum-degree bound in Lemma 5 and Sections 6.1–6.2, this proves:

Low-neighbor restriction. Every neighbor of a degree-five vertex has degree at least 7.

7. Overlap restriction for nonadjacent degree-five vertices

Proposition 8 (Overlap restriction; Proposition *O*). Two nonadjacent degree-five vertices in a minimal combined obstruction have at most three common neighbors. If they have three, both links are K_5.

Proof.

Attribution and extension. Botler–Fernandes–Gutiérrez Lemma 3.3 proves the corresponding implication in irreducible robust graphs: degrees at most six and a neighborhood union of size at most seven force both degrees to be five, an intersection of size three, and both links K_5. For two degree-five vertices, an intersection of at least three meets that union hypothesis. Its balanced reductions cannot simply exclude a minimal combined-S failure. The residual-block arguments in Sections 7.2–7.4 discharge that extra requirement; this is the precise distinction between the predecessor and the version needed here.

Write $C = N(x) \cap N(y)$. The two links are K_5 or K_5 minus one edge by Section 5. No assumption is made about other degrees.

7.1 Five common neighbors

Their common neighborhood A has five vertices and is K_5 or $K_5 - ab$. Delete x,y and $E(A)$; the cover increment is $|E(A)|$.

When $A = K_5$ on $\{a, b, c, d, e\}$, the six triangles xbc, xde, ybd, yce, abe, acd are disjoint. The cover increment ten is strictly smaller than twelve.

When $A = K_5 - ab$, omit abe from that list. The remaining five triangles exist and the cover increment nine is strictly smaller than ten.

7.2 Four common neighbors

Write $W = C$ and let p,q be the respective unique neighbors outside the common neighborhood. The graph on W is K_4 or K_4 minus one edge. It has two edge-disjoint perfect matchings, so four triangles, two at each center, form a packing.

Take $X = E(W) \cup \{xp, yq\}$. It meets all triangles at x,y and contains the noncentral edges of this four-packing. If W has five edges this is strict, with $|X| = 7$.

If $W = K_4$, the reduction is balanced four/eight and the residual has equality blocks with W independent. The vertex p has at least three neighbors in W, because the link at x misses at most one edge. Thus there is a page p f on some W-edge f. The graph on $\{x, y\} \cup W$ is $K_6 - xy$. Since xy and f are disjoint, it has a four-packing avoiding f. The local four-packing has no surviving residual edges: all its edges are incident with x, y or lie in W. A residual block meets the independent set W at most once, so the page contributes at most one surviving spoke. Thus the whole footprint is a star of size at most one, also below the K_5 two-edge bound. Lemma 2A appends the page and four-packing, giving $\nu(G) + 1$, a contradiction.

7.3 Three common neighbors, a missing edge inside C

If a common edge ac is absent, write $C = \{a, b, c\}$, $N(x) - C = \{p, q\}$, $N(y) - C = \{r, s\}$. Both links are precisely $K_5 - ac$; all edges required below therefore exist. The pairs p,q and r, s are disjoint, because C is exactly the common neighborhood.

Use $P = \{qab, bcr, xpa, xqc, yar, ycs\}$ and $X = E(qab) \cup E(bcr) \cup \{qc, pa, ar, cs, xp, ys\}$. The six triangles are disjoint and $|X| = 12$. It covers every triangle incident to x or y: xp handles the p-pages at x, qa,qb,qc handle its q-pages, ab,bc handle its C-triangles; the y side is analogous. Every noncentral P-edge lies in X. Thus the residual has equality blocks.

Now use the seven edge-disjoint triangles xpa, qab, pbc, xqc, yar, ysb, rsc. Their edges surviving in the residual are exactly pb,pc,sb,rs. Every pair of these except pb,sb requires an absent residual

edge to be in a common clique block: pb/pc requires bc, sb/rs requires br, pb/rs requires br, pc/sb requires bc, and pc/rs requires cr. Those edges were deleted. Hence each block contains at most two forbidden edges and the only possible pair shares b. The entire footprint therefore meets both hypotheses of Lemma 2A, which appends the seven-packing, contradiction.

7.4 Three common neighbors, a missing edge outside C

Suppose C is a triangle and at least one link is K_5 minus an edge; take that link to be the one at y. Let its missing edge be e. At least one endpoint of e lies outside C. Choose an edge $f = ac$ of C disjoint from e and let b be the third vertex.

Write $N(x) - C = \{p, q\}$. At least one of p,q joins all three C vertices; call it p. Relabel a,c if necessary so that qa exists. Then pb also exists.

The core on $\{y\} \cup N(y)$ is $K_6 - e$. Because e and f are disjoint, it has a four-packing P_0 avoiding f. Take $P = P_0 \cup \{xac, xpb\}$ and $X = E(G[N(y)]) \cup \{xp, xq, pb\}$. This is a balanced six/twelve reduction deleting x,y. The nine edges of the near-clique link at y cover all its triangles. Any x-triangle either contains p or q, hit by xp or xq, or has its other two vertices in C, whose edges are deleted. Noncentral packed edges all lie in X.

The residual has equality blocks and $N(y)$ is independent in it. Replace the two x-triangles by three triangles pac, xpb, xqa. Together with P_0 they form a seven-packing. Its surviving edges are pa,pc,qa: pb and ac were deleted. A residual block meets $N(y)$ at most once, so the whole footprint has at most two edges, with the only possible pair pa,qa, sharing a. Lemma 2A gives the contradiction.

All cases with a near-clique link are excluded, proving Proposition O.

8. The global peeling reduction and final local configuration

Lemma 9 (Final configuration and exclusion). A five-degenerate minimal combined obstruction forces the eight-vertex configuration described below, with arbitrary outside attachments, and this configuration cannot occur.

Proof.

We now exclude the minimal obstruction in the subgraph-closed class of all 5-degenerate graphs.

Let G be a minimal S failure among 5-degenerate graphs. Let L be its set of degree-five vertices. It is nonempty by degeneracy and Section 4. Section 6 makes L independent and gives degree at least 7 for all its neighbors.

The subgraph $G - L$ is nonempty and 5-degenerate. Choose u in it with at most five neighbors in $G - L$. Set $A = N(u) - L$, $D = d(u)$, $r = |N(u) \cap L|$. Then $|A| \leq 5$ and $r \geq D - 5$. Since u is outside L and $\delta(G) \geq 5$, $D \geq 6$, so $r \geq 1$. Thus u does have a neighbor in L, and the low-neighbor restriction gives $D \geq 7$ and $r \geq 2$.

For each v in $N(u) \cap L$, put $A_v = N(v) \cap A$. If u dominates v, $|A_v| = 4$; otherwise $|A_v| = 3$, and v has one further neighbor p_v outside $N[u]$. These descriptions use the precise K_5/K_5-minus-edge links. For distinct low vertices v,w, the sets A_v,A_w intersect in at most one vertex. Indeed two common A vertices together with u give at least three common neighbors. Proposition O excludes this if either link is a near-clique. If both links are cliques, both A sets have size four in an at-most-five-element set, so their intersection has size at least three and there are at least four common neighbors, again excluded.

There can be at most two subsets of an at-most-five-element set, each of size at least three, whose pairwise intersections have size at most one. To see this, two must each have size three, meet in one point and exhaust five points. Any third three-set would have total intersection size at least three with their union, whereas meeting each in at most one allows at most two. Thus $r = 2$, $D = 7$, $|A| = 5$, and both A sets are triples meeting in exactly one vertex.

Label $A = \{a, b, c, d, e\}$, $N(v) = \{u, p, a, b, c\}$, $N(w) = \{u, q, a, d, e\}$, $N(u) = \{v, w, a, b, c, d, e\}$. The low vertices v,w have near-clique links missing up,uq respectively. The vertices p,q lie outside $N[u]$ and are distinct: if they were equal, v,w would have three common neighbors u,a,p and Proposition O would exclude them. The sets $\{a, b, c\}$ and $\{a, d, e\}$ are triangles.

8.1 The four cross edges are forced

Suppose one of the four edges joining $\{b, c\}$ to $\{d, e\}$ is absent. Relabel within each pair so that ce is absent.

In the link at u, choose $M = \{vc, we, ab\}$, $Q = \{a, b, d\}$. Outside Q the vertices are v,c,w,e, whose only edges are vc and we: vw,ve,wc are absent by the displayed neighborhoods and ce is absent by assumption. Hence this is a balanced singleton three/six reduction.

The residual graph induced on $\{v, p, a, b, c\}$ is K_5 with vc and ab deleted; both pairs are nonedges of that residual. The other deleted edge we is outside this vertex set because p lies outside $N[u]$. This is impossible for the equality block classification. Thus all four cross edges exist and A is K_5.

8.2 The final eight-vertex graph B

Let B be the graph induced on $\{u, v, w, a, b, c, d, e\}$. It has $A = K_5$, u joined to all other seven vertices, v joined within B precisely to u,a,b,c, and w joined within B precisely to u,a,d,e. The only further neighbors of v,w in G are p,q respectively.

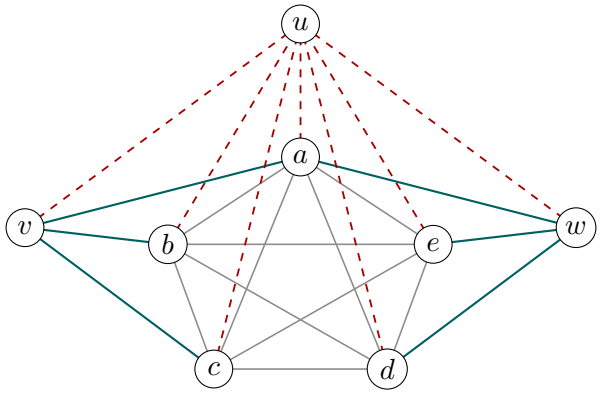


Figure 1: Exactly the 23 edges of B are drawn. Gray edges give $A = K_5$, dashed red edges join u to the other seven vertices, and teal edges give the two arms at v, w. Crossings are not additional vertices. The drawing excludes outside vertices: in G, v, w also have the respective neighbors p, q, and vertices of A may have unrestricted outside attachments. The displayed local graph does not bound those attachments.

A seven-packing of B avoiding ab is vua, vbc, wud, wae, uce, acd, bde. The two unused B edges are ub,ab. By swaps inside $\{b, c\}$, inside $\{d, e\}$, and exchange of the two arms together with v,w, there is a seven-packing avoiding each A-edge incident with a.

Pair-avoidance claim. For any two distinct A-edges f,g, $B - \{f, g\}$ has six edge-disjoint triangles.

If one edge is incident to a, take a seven-packing avoiding it and discard the possible triangle using the other edge. It remains to consider pairs among the six edges on $\{b, c, d, e\}$. Up to the same symmetries their types and witnesses are:

Avoided edges / type	Six edge-disjoint triangles
bc,de: two arm edges	vub, vac, wud, wae, uce, abd
bc,bd: arm edge and cross edge	vub, vac, wue, wad, ucd, abe
bd,ce: disjoint cross edges	vub, vac, wue, wad, ucd, abe
bd,be: adjacent cross edges	vua, vbc, wud, wae, uce, acd

These four cases exhaust pairs: an arm edge and any cross edge meet; two cross edges either meet or are disjoint. Every displayed triangle uses only B edges and the triangles in each row are pairwise edge-disjoint. Thus this is an ordinary finite certificate with an explicit symmetry reduction, not an appeal to an unperformed enumeration.

8.3 External-page augmentation closes the configuration

Delete u,v,w and every A-edge, obtaining K. The seven-packing of B appends to every K-packing. The fourteen edges $X = E(A) \cup \{uv, uw, vp, wq\}$ cover every triangle lost in the deletion. Triangles at u are inside B; triangles at v or w are inside B or use vp or wq; other lost triangles use an A-edge. Equality therefore forces $\nu(K) = \nu(G) - 7$, $\tau(K) = 2\nu(K)$, and K has K_4/K_5 triangle components. Since A is independent in K, each block meets A at most once.

Let F be the family of all triangles with an A-edge as base and apex outside B. If F contains two edge-disjoint members on bases f,g, use a six-packing of B avoiding f,g and append those two pages. The local six-packing has no edges in K. Each page contributes at most one surviving spoke to a residual block, because its two base endpoints lie in the independent set A. Thus the whole footprint is a star at the block's unique A vertex and has at most two edges, including for a K_5 block. Lemma 2A applies. This constructs $\nu(K) + 8 = \nu(G) + 1$ triangles, contradiction.

Thus F is pairwise edge-intersecting and has a cover of size at most two by the elementary intersecting-triangle lemma.

The following nine edges cover B: $\{ua, ub, ab, cd, ce, de, vc, wd, we\}$. The first six are the within-part edges of the $3 + 3$ partition $\{u, a, b\}/\{c, d, e\}$ of the K_6 core. The only remaining v-triangles use vc; the only remaining w-triangles use wd or we.

There are four separate cover contributions:

1. A minimum K-cover handles all triangles surviving in K.
2. The displayed nine-edge cover handles all triangles wholly in B.
3. An F-cover of size at most two handles all triangles with an A-edge and apex outside B.
4. The two edges vp,wq handle every remaining triangle incident with the deleted low vertices v,w.

To check exhaustiveness, a lost triangle either meets $\{u, v, w\}$ or uses a deleted A-edge. Any triangle at u is wholly in B because $N(u) = \{v, w\} \cup A$. A triangle at v either lies in B or uses its unique outside-neighbor edge vp; similarly w and wq. A triangle using an A-edge either has its third vertex in B, or is an external page in F. All graph edges chosen in the F-cover are actual

edges of G. Contributions can overlap, which only reduces the size of their union; no disjointness is needed for the upper bound. Thus a minimum K-cover plus the other contributions has size at most $\tau(K) + 9 + 2 + 2 = \tau(K) + 13$, contradicting $\tau(G) = \tau(K) + 14$.

For the inequality alone, the earlier seven/fourteen reducing tuple already contradicts irreducibility, as reconstructed in the short credited proof. The saving from fourteen to thirteen and the argument for arbitrary outside pages are retained here to exclude equality obstructions.

Therefore the minimal failure cannot exist.

9. Conclusion, converse and scope

The class of finite 5-degenerate graphs is closed under every vertex and edge deletion used above. If it contained a failure of S, choose one with the fewest edges and discard isolates. Sections 1–5 give the minimal-obstruction premises. Sections 6–8 exclude it. Hence every graph in this class satisfies the inequality and the asserted necessity at equality.

Conversely, packing and covering numbers add over triangle components because their edge supports are disjoint. K_4 has $(\nu, \tau) = (1, 2)$, K_5 has (2,4), and triangle-free edges contribute zero. Thus the stated component condition implies equality, including the vacuous triangle-free case.

This proves the main theorem by ordinary finite graph arguments. The public finite-certificate checkers independently verify displayed local certificates and the supplied small-link coverage; they do not enumerate arbitrary external attachments and are not substitutes for the induction or page arguments above. The formal-verification account appears in Section 10; the ordinary argument and finite checks have distinct evidentiary roles. No external human endorsement or established historical priority is claimed.

For an individual graph independently known to be 5-degenerate, this theorem supplies Tuza's inequality and the stated equality characterization. No unrestricted restoration, strictness, OCT-five, or density-class theorem is established here. Here OCT-five means the class of graphs made bipartite by deleting at most five vertices; no assertion for that entire class follows from the degeneracy theorem.

10. Formal verification

The exact combined result is proved by `Tuza06.five_degenerate_combined`. Its only substantive hypothesis is five-degeneracy on an arbitrary finite simple graph. The formal definitions use the full graph triangle family, edge-disjoint packing, graph-edge covering, induced-subset degeneracy and literal shared-edge component supports. The component/block equivalence and the equality converse are proved. Unrelated conditional interfaces in reused files are not premises of the main theorem.

The existing verification uses Lean 4.33.1, pinned Mathlib and eight other pinned packages. It rebuilt all 280 mathematical modules, audited 618 public theorem types and transitive axiom closures, and replayed all 280 fresh local module objects. Only `propext`, `Classical.choice` and `Quot.sound` occur. Compiled pinned dependencies were reused, and replay used the same standard Lean kernel; it was neither a cold Mathlib rebuild nor an independent kernel implementation. Appendix C preserves the exact target, declaration coverage, source identities and execution provenance; the public verification and reproduction record gives the commands and limits. This editorial revision does not repeat those Lean runs.

Appendix A. A short credited proof of the inequality

A.1 Published inputs and exact scope

Puleo, [2], Definition 2.1 and Lemma 2.2, supplies the reducible-set framework. Lemma 2.6 gives robustness for a minimal inequality counterexample. In a robust graph with no reducible set, Lemma 2.7(a),(*b*), equivalently Propositions 5.1 and 5.2, implies that degree-five links are K_5 or K_5 minus one edge, and vertices of degree at most six form an independent set. Here robustness means that every component of every vertex link has at least five vertices. The complement condition in Proposition 5.1 is part of its hypothesis.

Botler–Fernandes–Gutiérrez, [3], Lemma 3.3, applies to irreducible robust graphs. For two degree-five vertices with at least three common neighbors it forces exactly three common neighbors and both links K_5. This is the published antecedent used below. Their theorem concerns treewidth; it is this local lemma that is used here.

A.2 Reduction accounting

Use the reducible-set witnesses U, P, X specified after Lemma 3A. Its packing and cover inequalities give $\nu(K) + r \leq \nu(G)$ and $\tau(G) \leq \tau(K) + |X|$. Therefore $\tau(K) \leq 2\nu(K)$ and $|X| \leq 2r$ already imply $\tau(G) \leq 2\nu(G)$. Equality of the cover bound is sufficient for this inequality-only induction; the residual equality conclusion of Lemma 3A is needed only for the combined-obstruction proof.

Assume for a contradiction that a 5-degenerate counterexample exists. Choose one with the fewest edges, discard isolated vertices, and, if necessary, minimize the number of vertices among equal edge counts. Every proper subgraph satisfies the inequality: it is still 5-degenerate. Hence the published robustness lemma applies, and no reducible vertex set exists. Denote this graph by G. Robustness gives minimum degree at least five. The published low-degree facts above give:

- every degree-five vertex has link K_5 or K_5 minus one edge;
- degree-five vertices are independent and all their neighbors have degree at least seven;
- two degree-five vertices have at most three common neighbors, and if they have three then both links are cliques.

These are uses of inequality irreducibility. They are not asserted for a combined equality obstruction without further proof.

A.3 Degeneracy peeling

Let L be the set of degree-five vertices. It is nonempty by 5-degeneracy and minimum degree five. It is independent. $G - L$ is nonempty because a vertex in L has neighbors outside L. Choose u in $G - L$ with degree at most five in $G - L$, and set

$$A = N(u) - L, r = |N(u) \cap L|, D = d_G(u)$$

Then $|A| \leq 5$, $r = D - |A| \geq D - 5$, and $D \geq 6$ because u is outside L. Thus $r \geq 1$. A neighbor in L forces $D \geq 7$, so $r \geq 2$. The second use of degeneracy bounds degree in $G - L$, not degree in G.

For each v in $N(u) \cap L$, let $A_v = N(v) \cap A$. Because L is independent, every neighbor of v other than u lies outside L. If the link at v is a clique, u dominates v and $|A_v| = 4$. If that link misses

an edge not incident with u, the same conclusion holds. If its missing edge is up_v, then $|A_v| = 3$ and p_v is the unique neighbor of v outside $N[u]$. These exhaust the cases. In all cases $|A_v| \geq 3$.

For distinct low neighbors v,w of u, suppose $|A_v \cap A_w| \geq 2$. The common neighbors of v,w include u and those two vertices, so the published overlap lemma makes both links cliques and gives exactly three common neighbors. But then A_v and A_w are both four-subsets of the at-most-five-element set A, so they meet in at least three vertices. Together with u this gives at least four common neighbors, a contradiction. Hence the sets A_v have pairwise intersections at most one.

There are at most two subsets of a set of size at most five, each of size at least three and pairwise intersecting in at most one element. Indeed, any two must be three-sets meeting in one point and covering all five points; a third three-set cannot meet each of the first two in at most one point. Since $r \geq 2$, it follows that $r = 2$, $D = 7$, $|A| = 5$, and the two sets A_v,A_w are triples meeting in one point.

Label

$$A = \{a, b, c, d, e\},$$
$$N(v) = \{u, p, a, b, c\},$$
$$N(w) = \{u, q, a, d, e\},$$
$$N(u) = \{v, w, a, b, c, d, e\}$$

Both low links are near-cliques, missing up and uq respectively. The vertices p,q lie outside $N[u]$. They are distinct, because otherwise v,w would have common neighbors u,a,p while their links were not cliques. The sets $\{a, b, c\}$ and $\{a, d, e\}$ are triangles, and p joins a,b,c while q joins a,d,e.

A.4 The final reduction

If a cross edge between $\{b, c\}$ and $\{d, e\}$ is missing, swap labels within the two pairs so that ce is absent. In the link at u use the matching

$$M = \{vc, we, ab\}$$

and $Q = \{a, b, d\}$. The link after removing Q has vertices v,c,w,e and only edges vc,we: the other possible edges are absent from the displayed neighborhoods or by ce being missing. Therefore the three triangles

$$uvc, uwe, uab$$

and the six edges

$$X = \{vc, we, ab, ua, ub, ud\}$$

make $\{u\}$ reducible. Every u-triangle either has a neighbor in Q, or uses a matching edge outside Q. The noncentral edges of all three packed triangles lie in X. This contradicts irreducibility. Thus all four cross edges exist and A is K_5.

Let B be the local graph on $\{u, v, w, a, b, c, d, e\}$. Its edges are the ten A-edges, all seven edges from u to the other local vertices, va,vb,vc, and wa,wd,we. The following seven triangles exist and are pairwise edge-disjoint:

$$vua, vbc, wud, wae, uce, acd, bde$$

Take $U = \{u, v, w\}$ and

$$X = E(A) \cup \{uv, uw, vp, wq\}$$

There are fourteen edges in X. Every triangle at u is wholly in B and either uses uv, uw, or an A-edge. Every triangle at v or w either uses the corresponding outside-neighbor edge vp or wq, or is local; its local case uses uv, uw, or an A-edge. Thus X meets every triangle incident with U. Every edge of the displayed seven-packing outside U is an A-edge and belongs to X. Hence this is a balanced seven/fourteen reduction, again contradicting irreducibility.

No minimal inequality counterexample exists, proving $\tau(G) \leq 2\nu(G)$ for every finite simple 5-degenerate graph G. Empty and triangle-free graphs have both parameters zero and satisfy the assertion directly.

Appendix B. The small-link proof used by the formal development

The formal development unifies the diamond and four-cycle arguments of Sections 4.3–4.4 as follows. It changes the organization, not the theorem or attribution boundary.

Let $C = \{a, b, c, d\}$ be closed in the link at v, with ab absent and ac,ad,bc,bd present; cd is unrestricted. Delete only the four v-spokes into C and rim edges ac,bd. The two triangles vac,vbd append to every residual packing, and ac,bd,vc,vd cover every lost triangle. Balanced two/four accounting gives residual equality blocks and the exact increment $\nu(G) = \nu(H) + 2$. Closure restricts only link edges; arbitrary external pages remain allowed at this point.

For an external ac-page with apex x, combine that page with vad,vbc. Their surviving footprint is ad,bc,xa,xc. Since ac is absent in H, the two terminals a,c are independent; each residual clique meets at most one terminal and sees at most two forbidden incident edges. The whole footprint is a star of size at most two, so Lemma 2A gives $\nu(H) + 3$, a contradiction. The bd case is the same explicit relabeling with terminals b,d. Thus every lost triangle using ac or bd is local. The remaining lost hub triangles are covered by vc,vd. Every rim triangle must contain c,d because ab is absent. If cd exists, adding cd yields a three-edge lost-triangle cover; if cd is absent, there is no rim triangle and vc,vd suffice. Either contradicts the exact four-edge increment. This excludes both local link types.

Appendix C. Declaration coverage and verification supplement

The exact public target is `Tuza06.five_degenerate_combined`:

```
@Tuza06.five_degenerate_combined :
  ∀ {V : Type u} [Fintype V] [DecidableEq V] {G : SimpleGraph V},
    Tuza06.Degenerate G 5 →
      Tuza.tau (Tuza.triangles G) ≤ 2 * Tuza.nu (Tuza.triangles G) ∧
      (Tuza.tau (Tuza.triangles G) = 2 * Tuza.nu (Tuza.triangles G) ↔
        Tuza06.ComponentCliques G)
```

The universe binder above is displayed using the conventional name `u`; generated printer names may differ. `triangles G` is the full `G.cliqueFinset 3`. `nu` counts edge-disjoint triangle families. `tau` uses actual two-vertex edge covers; `Tuza.graph_min_cover` proves attainment by graph edges. `Degenerate` uses the usual induced-subset definition. `ComponentCliques` uses reflexive transitive closure of shared-edge triangle adjacency and literal edge support. The theorem has no unproved external mathematical inputs. Its only substantive hypothesis is `Degenerate G 5`; the ambient finite-type and decidable-equality instances are shown explicitly.

The original target `CombinedStatement G` is proved by `five_degenerate_statement`. Definition-stage comments in the source record the target before completion; the final theorem and current proof status supersede those historical comments. The four reused `Tuza` modules contain unrelated conditional interfaces, but their unproved mathematical inputs are not premises of this theorem.

Paper result or step	Actual Lean declaration(s), in namespace `Tuza06` unless stated otherwise
Theorem 1, both conclusions	`five_degenerate_combined`
Lemma 2A, block-packing compatibility	Ordinary blockwise formulation from Lemma 2; uses existing `k5_avoid_two`, `Tuza.k4_face_avoiding` and `blocks_combine_packings/blocks_nu`. No separately named exact general Lean lemma is claimed. The two-page specialization is covered by `blocks_avoid_two_pages` with its explicit independent-base hypotheses.
Lemma 3A, balanced reduction	`packing_extension`; `cover_extension`; `balanced_accounting`; `strict_accounting`; the classification step uses actual minimality through `minimal_balanced_reduction`.
Separate inequality and equality	`five_degenerate_inequality`; `five_degenerate_equality`
Original target and block equivalence	`five_degenerate_statement`; `componentCliques_iff_decomposition`
Equality converse	`componentCliques_equality`
Lemma 2, clique parameters and avoidance	`k4_parameters`; `k5_parameters`; `k5_avoid_two`; `Tuza.k4_face_avoiding`
Intersecting triangle families	`Tuza.intersecting_cover`; `Tuza.intersecting_k4_structure`
Lemma 3, minimality and criticality	`exists_minimal_obstruction`; `minimal_tight_critical`
Lemma 4, every edge in two triangles	`minimal_edge_codegree`
Lemma 5, closed link size	`minimal_closed_link_size`

Paper result or step	Actual Lean declaration(s), in namespace `Tuza06` unless stated otherwise
Lemma 6, all degree-five links	`minimal_degree_five_configuration`
Lemma 7, low adjacency and high neighbors	`minimal_degree_five_independent`; `minimal_degree_five_neighbors_high`
Proposition 8, complete overlap restrictions	`minimal_degree_five_common_le_three`; `minimal_three_common_cliques`
Lemma 9, second degeneracy and labels	`minimal_two_arm_peeling`; `minimal_final_arm_data`
Actual final configuration	`final_configuration_of_arm_data`
Arbitrary pages and strict final cover	`minimal_no_final_configuration`
Auxiliary four-degenerate combined theorem	`four_degenerate_combined`

The prescribed-edge minimum-cover observation in Lemma 2 is an elementary ordinary detail used in Section 4.4. It is not separately exported as a general K_5 cover declaration. The formal development instead uses the unified small-link argument in Appendix B; the complete formal theorem includes that route.

The accepted source inventory covers **280 mathematical modules**, all in the main entry point's import closure. The source of `FiveDegenerate.lean` has SHA-256 `978ebe0431fde9ecf9598538c6c01e06bcd9ee363bd5ee03a53b76c43143fadf`. All 280 files are preserved byte-for-byte. Lean **4.33.1** and Mathlib revision `0df444a360eaa60ab8c11dca51a86af692955474`, together with all nine package revisions, are pinned. Complete theorem types and transitive-axiom audits cover **618 public theorem declarations**; the only reported axioms are `propext`, `Classical.choice`, and `Quot.sound`. There is no `sorry`, custom mathematical axiom, hidden external theorem premise, or native proof shortcut in the mathematical closure.

The original source verification rebuilt every mathematical module in a fresh project workspace and replayed all **280 fresh module objects** serially through the standard Lean kernel, including private proof constants. The original publication verification separately rebuilt and replayed the exact public source tree; its outcome, commands and evidence are recorded in the portable reproduction record. Reused compiled dependencies were separately pinned and checked. This is **not a cold Mathlib rebuild**. The replay adapter retains the upstream LeanChecker replay routine; its unsafe IO/runtime-memory wrapper is outside the mathematical import closure. Replay is additional checking by the same standard Lean kernel, **not an independently implemented kernel**. Upstream source provenance and license accompany the tool.

The independent Python checks validate supplied local finite certificates, five-link coverage and overlap witnesses. They do not enumerate arbitrary outside attachments and are not premises of the formal theorem. The symbolic page and compatibility arguments in Section 8 discharge that unbounded obligation. Neither kernel checking nor internal review certifies novelty, human specialist endorsement, or broader conjectures.

The public research archive supplies exact Lean sources, full public proof records, relevant finite certificates/checkers, attribution and provenance, selected reviews, historical ordinary proofs and their limitations, source-to-publication mapping, and reproduction commands. These records distinguish earlier partial status from the completed theorem. Operational instructions and private review archives are excluded.

Other formal changes are organizational: finite label coverage is proved explicitly; all reductions are instantiated from actual neighborhoods; outside pages and residual block compatibility are handled symbolically. Their exact declarations, dependencies and completion status belong to the

formal status record, rather than to an inference from this manuscript.

The source verification and its 280-module rebuild/replay are historical mathematical evidence, retained unchanged. This editorial release checks the accepted source hashes and revises exposition, literature positioning and presentation; it does not report a new Lean build or replay. The current editorial artifact record distinguishes the revised PDF from the earlier verified paper hash.

Data availability and report status

This is an explicitly AI-developed mathematical research experiment and technical report. The public archive [7] provides the ordinary proof records, exact Lean sources, finite certificates, reproduction instructions and internal review records. The title-page disclosure identifies the AI development and the human participant's coordination role. Internal AI reviews and kernel checking do not constitute external specialist peer review or establish historical priority. Published results used here retain their human authors' attribution.

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